

Reproof of the Mahar–Willner Doubling Lemmas

(Signed Cell Extension, Wronskian Zero Motion, and Zero Truncation)

BVE research project — proof revision author

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Abstract

We give an author’s revised proof of the two structural lemmas associated with Mahar–Willner [1] in their solution of the extremal problem for λ_2/λ_1 of $-y'' = \lambda \rho y$ (Dirichlet, $a \leq \rho \leq 1$): Lemma 1 (the *cell-extension* identity $\lambda_{2n}(\rho_n)/\lambda_n(\rho_n) = \lambda_2(\rho_0)/\lambda_1(\rho_0)$ for the $1/n$ -periodic “blow-up” ρ_n of an arbitrary ρ_0) and Lemma 2 (the *zero-truncation* induction establishing $\mu_{2n,n}(a) = \mu(a)$, $\nu_{2n,n}(a) = \nu(a)$). Here μ, ν are defined as the first-pair infimum and supremum; the doubling lemmas do not determine their values. The proof uses classical Sturm oscillation and comparison and compact self-adjoint spectral theory. We include the common-zero equality branch, prove zero motion by a parameter Wronskian, and justify passage from finite step densities to piecewise continuous and bounded measurable densities. The explicit first-pair value requires the separate variational theorem in `SL_ratio_proof.tex`, not merely its balanced-candidate calculation. This is a proof submission by the revision author, not an independent acceptance report.

1 Setting and notation

Initially, ρ is a finite step density on $I = [-1/2, 1/2]$ with $a \leq \rho \leq 1$ almost everywhere, where $0 < a \leq 1$, and $\mathcal{P}_a$ is this class. Write $\mathcal{Q}_a$ for all $L^\infty(I)$ densities with the same bounds and $\mathcal{PC}_a$ for the piecewise continuous subclass. Values at jump points have no spectral effect. Consider

$$-y'' = \lambda \rho(x) y, \quad y(-1/2) = y(1/2) = 0, \quad (1)$$

with eigenvalues $0 < \lambda_1(\rho) < \lambda_2(\rho) < \dots$ (simplicity and discreteness are classical). Write

$$\mu(a) := \inf_{\rho \in \mathcal{P}_a} \frac{\lambda_2}{\lambda_1}, \quad \nu(a) := \sup_{\rho \in \mathcal{P}_a} \frac{\lambda_2}{\lambda_1}, \quad \mu_{2n,n}(a) := \inf_{\rho \in \mathcal{P}_a} \frac{\lambda_{2n}}{\lambda_n}, \quad \nu_{2n,n}(a) := \sup_{\rho \in \mathcal{P}_a} \frac{\lambda_{2n}}{\lambda_n}.$$

The eigenfunction of λ_j is denoted y_j ; it is normalized arbitrarily, and we always take $y'_j(-1/2) \neq 0$ (true up to rescaling). For measurable densities the equation holds almost everywhere and $y_j \in W^{2,\infty}(I) \subset C^1(I)$. In particular both y_j and y'_j match across every interface. The variational bounds $(j\pi)^2 \leq \lambda_j \leq (j\pi)^2/a$ show that all fixed-index extrema in the display are finite.

Proposition 1.1 (density classes and spectral continuity). *If $\rho_l, \rho \in \mathcal{Q}_a$ and $\|\rho_l - \rho\|_1 \rightarrow 0$, then $\lambda_j(\rho_l) \rightarrow \lambda_j(\rho)$ for every fixed j . Each fixed-index ratio has the same supremum and infimum over $\mathcal{P}_a$, $\mathcal{PC}_a$ and $\mathcal{Q}_a$.*

Proof. On $H = H_0^1(I)$ with inner product $\int u'v'$, define the compact positive self-adjoint operator $(T_\rho u, v)_H = \int \rho uv$. Since $|I| = 1$, $\|u\|_\infty \leq \|u'\|_2$, so $\|T_{\rho_l} - T_\rho\| \leq \|\rho_l - \rho\|_1$. Its positive eigenvalues are $1/\lambda_j(\rho)$; the compact self-adjoint min–max principle proves fixed-index continuity. Approximate ρ by its averages on dyadic interval partitions. These finite step densities preserve $[a, 1]$ and converge

in L^1 . Indeed, interval averaging is an L^1 contraction and converges for continuous functions, which are dense in L^1 . The asserted equalities follow from continuity and the inclusions $\mathcal{P}_a \subset \mathcal{PC}_a \subset \mathcal{Q}_a$. This argument does not assert uniform convergence in the eigenvalue index. $\square$

Preliminary facts. We shall use the following classical results; proofs are sketched only where the formulation is non-standard.

Proposition 1.2 (affine invariance of ratios). *Let $L(x) = cx + d$ with $c \neq 0$ map $[x_0, x_1]$ onto $[-1/2, 1/2]$. For every index j , the Dirichlet eigenvalue on $[x_0, x_1]$ with density $\rho \circ L$ is $c^2 \lambda_j(\rho)$. All eigenvalue ratios are invariant, including when $c < 0$.*

Proof. If $v = y \circ L$, then $-v'' = c^2 \lambda(\rho \circ L)v$. Composition with L is a bijection between Dirichlet solutions and preserves the number of interior zeros; this identifies the index. The same statement applies between any two finite nondegenerate intervals. $\square$

Proposition 1.3 (Sturm oscillation). *The j -th eigenfunction y_j of (1) has exactly $j - 1$ (interior) zeros in $(-1/2, 1/2)$, all simple.*

Proposition 1.4 (Sturm interlacing). *If $j < k$ and $z_1 < \dots < z_{j-1}$ are the zeros of y_j , then y_k has at least one zero in each of the j intervals $(-1/2, z_1), (z_1, z_2), \dots, (z_{j-1}, 1/2)$.*

The backbone of the truncation argument is the following elementary monotonicity of the zeros of an initial-value problem (IVP).

Proposition 1.5 (zero motion). *Fix $c \neq 0$ and let $y(\cdot; \lambda)$ be the solution of*

$$y'' + \lambda \rho(x)y = 0, \quad y(-1/2; \lambda) = 0, \quad y'(-1/2; \lambda) = c.$$

For $j \geq 1$ let $z_j(\lambda)$ be the j -th zero of $y(\cdot; \lambda)$ to the right of $-1/2$, whenever it exists. Then each z_j is continuous and strictly decreasing in λ . The analogous statement holds for the IVP posed at $x = 1/2$: the j -th zero to the left of $1/2$ is strictly increasing in λ .

Proof. Put $\ell = -1/2$. The Volterra equation

$$y(x; \lambda) = c(x - \ell) - \lambda \int_{\ell}^x (x - t)\rho(t)y(t; \lambda) dt$$

gives continuously differentiable parameter dependence, also for bounded measurable ρ . Write $v = \partial_{\lambda} y$; then $v'' + \lambda \rho v = -\rho y$ and $v(\ell) = v'(\ell) = 0$. The absolutely continuous Wronskian satisfies

$$(y'v - yv')' = \rho y^2 \quad \text{a.e.}, \quad y'(z_j)v(z_j) = \int_{\ell}^{z_j} \rho y^2.$$

Every zero is simple by uniqueness of the IVP. The implicit function theorem therefore gives

$$\frac{dz_j}{d\lambda} = -\frac{v(z_j)}{y'(z_j)} = -\frac{\int_{-1/2}^{z_j} \rho(x)y(x; \lambda)^2 dx}{y'(z_j; \lambda)^2} < 0. \quad (2)$$

Simplicity prevents collisions, so the local branches preserve their ordered indices. At an endpoint one may extend ρ by any positive bounded density to interpret the crossing. From an IVP fixed at $1/2$, the same identity integrated backwards gives instead

$$\frac{dz_j}{d\lambda} = \frac{\int_{z_j}^{1/2} \rho(x)y(x; \lambda)^2 dx}{y'(z_j; \lambda)^2} > 0.$$

All identities are integrated identities for absolutely continuous functions, so there are no missing density-interface terms. $\square$

An equivalent formulation, used repeatedly below, is:

Corollary 1.6. *Let $\bar{y}(\cdot; \lambda)$ be an IVP solution as above and let $x_* \in (-1/2, 1/2)$ be a fixed point which is not a zero of $\bar{y}(\cdot; \lambda_0)$. As λ increases (resp. decreases) from λ_0 , the value $\bar{y}(x_*; \lambda)$ changes sign exactly when a zero of $\bar{y}$ crosses x_* ; zeros cross one at a time, and the count of zeros of $\bar{y}$ strictly between its fixed shooting endpoint and x_* changes by ± 1 at each crossing. For an arbitrary interval with endpoint x_* , this count statement requires that no zero simultaneously crosses the other endpoint.*

Lemma 1.7 (shooting count on a truncated interval). *On an interval $J = (b, c)$, let $e_1 < e_2 < \dots$ be its Dirichlet eigenvalues with the restricted density. For a nonzero IVP solution based at either endpoint at $\lambda > 0$, let N be the number of its zeros strictly inside J . If it vanishes at the other endpoint, then $\lambda = e_{N+1}$. Otherwise*

$$e_N < \lambda < e_{N+1}, \quad e_0 := 0.$$

In particular $\lambda \leq e_{N+1}$ always, and $\lambda > e_N$ if $N \geq 1$.

Proof. At $\lambda = 0$ the IVP solution is a nonzero linear function with no interior zero. The zero count is locally constant unless the opposite endpoint is a zero: zeros are simple, cannot collide, and cannot pass through the initial endpoint, whose derivative is fixed and nonzero. Such endpoint zeros occur precisely at the Dirichlet eigenvalues. By (2), exactly one zero enters J as λ increases through each eigenvalue. At e_j , oscillation gives $j - 1$ strictly interior zeros. These facts give the asserted intervals and equality cases; reflection gives the opposite shooting direction. $\square$

2 Lemma 1: the cell extension

Lemma 2.1 (cell extension). *For every $\rho_0 \in \mathcal{P}_a$, every integer $n \geq 1$, and the $1/n$ -periodic density*

$$\rho_n(x) := \rho_0\left(n\left(x + \frac{1}{2}\right) - \frac{1}{2}\right), \quad -\frac{1}{2} \leq x \leq -\frac{1}{2} + \frac{1}{n}, \quad \text{extended with period } 1/n,$$

one has $\rho_n \in \mathcal{P}_a$ and

$$\frac{\lambda_{2n}(\rho_n)}{\lambda_n(\rho_n)} = \frac{\lambda_2(\rho_0)}{\lambda_1(\rho_0)}. \quad (3)$$

Proof. Fix ρ_0 and let y_1, y_2 be its first two eigenfunctions. Define the nonzero, signed transmission factors

$$b_i := \frac{y'_i(1/2)}{y'_i(-1/2)}, \quad i = 1, 2.$$

Zero counts show $b_1 < 0$ and $b_2 > 0$: the second eigenfunction's endpoint derivatives have the same sign. Their magnitudes need not equal one or each other.

Set $x_j := -1/2 + j/n$ and, for $x \in [x_j, x_{j+1}]$, $u_j(x) := n(x - x_j) - 1/2 \in [-1/2, 1/2]$. Define

$$\tilde{y}(x) := b_1^j y_1(u_j(x)), \quad \tilde{f}(x) := b_2^j y_2(u_j(x)), \quad x \in [x_j, x_{j+1}], \quad j = 0, \dots, n-1.$$

Both are C^1 across every cell boundary x_{j+1} : continuity holds because $y_1(\pm 1/2) = y_2(\pm 1/2) = 0$, and the slopes match because

$$nb_i^j y'_i(1/2) = nb_i^{j+1} y'_i(-1/2), \quad i = 1, 2.$$

On cell j one has $\tilde{y}'' = -n^2 \lambda_1 \rho_n \tilde{y}$ and $\tilde{f}'' = -n^2 \lambda_2 \rho_n \tilde{f}$, so both are eigenfunctions of (1) with $\rho = \rho_n$, of eigenvalues $n^2 \lambda_1$ and $n^2 \lambda_2$. Their boundary values at $\pm 1/2$ vanish.

Zeros: $\tilde{y}$ vanishes exactly at the cell boundaries, i.e. at $n - 1$ interior points, hence it is the n -th eigenfunction (Proposition 1.3) and $\lambda_n(\rho_n) = n^2 \lambda_1(\rho_0)$. The function $\tilde{f}$ vanishes at the $n - 1$ cell boundaries and at one point inside each cell (the image of the interior zero of y_2), i.e. at $2n - 1$ interior points, hence it is the $2n$ -th eigenfunction and $\lambda_{2n}(\rho_n) = n^2 \lambda_2(\rho_0)$. Dividing gives (3). $\square$

Corollary 2.2. $\mu_{2n,n}(a) \leq \mu(a)$ and $\nu_{2n,n}(a) \geq \nu(a)$ for all $n \geq 1$.

Proof. For any ρ_0 , (3) says that the value $\lambda_2(\rho_0)/\lambda_1(\rho_0)$ is attained by the ratio λ_{2n}/λ_n at $\rho_n \in \mathcal{P}_a$. Taking the supremum (resp. infimum) over ρ_0 gives $\nu_{2n,n}(a) \geq \nu(a)$ (resp. $\mu_{2n,n}(a) \leq \mu(a)$). $\square$

3 Lemma 2: the zero-truncation induction

Let $k \geq 1$, $m = k + 1$, and $\rho \in \mathcal{P}_a$ be arbitrary. Write z_0 for the rightmost interior zero of y_m , and enumerate the interior zeros of y_{2m} unambiguously as

$$z_1 < z_2 < \cdots < z_{2k+1}.$$

Thus z_{2k} is its second-from-right interior zero. Comparison gives at least k zeros of y_{2m} strictly inside $L = (-1/2, z_0)$ and at least one strictly inside $J = (z_0, 1/2)$; in particular $z_0 < z_{2k+1}$. Write e_j^L, e_j^J for the spectra of these two restricted problems. Since y_m has $k - 1$ zeros inside L and none inside J ,

$$e_k^L = e_1^J = \lambda_m(\rho). \quad (4)$$

Affine normalization preserves the density bounds and every ratio, so all extrema defined on the unit interval apply to these restricted spectra.

3.1 The maximum case

Lemma 3.1 (truncation, max). *For every $\rho \in \mathcal{P}_a$ and $k \geq 1$,*

$$\frac{\lambda_{2(k+1)}(\rho)}{\lambda_{k+1}(\rho)} \leq \max\{\nu_{2k,k}(a), \nu(a)\}.$$

Proof. *Case A:* $z_0 \leq z_{2k}$. Let j be the number of zeros of y_{2m} strictly inside L . Then $k \leq j \leq 2k - 1$. Apply Lemma 1.7 to the left-based solution at λ_{2m} , which is a nonzero multiple of y_{2m} . It gives

$$\lambda_{2m} \leq e_{j+1}^L \leq e_{2k}^L, \quad \frac{\lambda_{2m}}{\lambda_m} \leq \frac{e_{2k}^L}{e_k^L} \leq \nu_{2k,k}(a).$$

This includes any common zero at z_0 : in that event $\lambda_{2m} = e_{j+1}^L$. In particular $z_0 = z_{2k}$ gives $j = 2k - 1$ and exact equality with e_{2k}^L .

Case B: $z_{2k} < z_0$. Here $z_{2k} < z_0 < z_{2k+1}$, so $y_{2m}(z_0) \neq 0$ and exactly one zero lies inside J . Right-based shooting and Lemma 1.7 give $\lambda_{2m} < e_2^J$. Thus

$$\frac{\lambda_{2m}}{\lambda_m} \leq \frac{e_2^J}{e_1^J} \leq \nu(a).$$

The two cases exhaust all positions of z_0 . $\square$

Theorem 3.2 (max doubling identity). $\nu_{2n,n}(a) = \nu(a)$ for every $n \geq 1$.

Proof. The base case is $n = 1$, namely $\nu_{2,1} = \nu$ by definition. Assume the assertion for $n = k$. Lemma 3.1, for each density separately, gives $\lambda_{2(k+1)}/\lambda_{k+1} \leq \nu(a)$. Taking the supremum proves the upper bound for $n = k + 1$; Corollary 2.2 supplies the reverse bound. $\square$

3.2 The minimum case

Lemma 3.3 (truncation, min). *For every $\rho \in \mathcal{P}_a$ and $k \geq 1$,*

$$\frac{\lambda_{2(k+1)}(\rho)}{\lambda_{k+1}(\rho)} \geq \min\{\mu_{2k,k}(a), \mu(a)\}.$$

Proof. There are three cases, including equality.

Case A: $z_0 < z_{2k}$. The number c_0 of zeros of y_{2m} strictly inside J is at least two. Lemma 1.7 gives $\lambda_{2m} > e_{c_0}^J \geq e_2^J$, whether or not z_0 itself is an earlier zero of y_{2m} . Hence

$$\frac{\lambda_{2m}}{\lambda_m} \geq \frac{e_2^J}{e_1^J} \geq \mu(a).$$

Case B: $z_0 = z_{2k}$ (common zero). Restrict both eigenfunctions to J . The first has no interior zero and the second has exactly one. They satisfy both Dirichlet boundary conditions, so

$$e_1^J = \lambda_m, \quad e_2^J = \lambda_{2m}, \quad \frac{\lambda_{2m}}{\lambda_m} = \frac{e_2^J}{e_1^J} \geq \mu(a).$$

No shooting perturbation is needed. The same restrictions on L also have indices k and $2k$, respectively.

Case C: $z_0 < z_{2k}$. Now all of $z_1, \dots, z_{2k}$ lie inside L and $y_{2m}(z_0) \neq 0$. Thus Lemma 1.7 gives $\lambda_{2m} > e_{2k}^L$, and

$$\frac{\lambda_{2m}}{\lambda_m} \geq \frac{e_{2k}^L}{e_k^L} \geq \mu_{2k,k}(a).$$

These inequalities imply the claim. $\square$

Remark 3.4 (constant-density equality case). *For $\rho \equiv d \in [a, 1]$ and every $m \geq 2$, the rightmost zero of y_m is $z_0 = 1/2 - 1/m = z_{2m-2} = z_{2k}$ for y_{2m} . The ratio is 4; minimum Case B and maximum Case A both apply. Thus $c_0 = 1$ does not imply $z_{2k} < z_0$.*

Theorem 3.5 (min doubling identity). $\mu_{2n,n}(a) = \mu(a)$ for every $n \geq 1$.

Proof. The base case is $n = 1$, namely $\mu_{2,1} = \mu$ by definition. Assuming the assertion for $n = k$, apply Lemma 3.3 and take the infimum to obtain $\mu_{2(k+1),k+1} \geq \mu$. Corollary 2.2 supplies the reverse bound. $\square$

Corollary 3.6 (extension to measurable densities). *Both doubling identities hold with the extrema taken over $\mathcal{PC}_a$ or $\mathcal{Q}_a$ as well.*

Proof. For each fixed n , Proposition 1.1 identifies the extrema over all three classes. There is no interchange of a limit with an unbounded eigenvalue index. $\square$

4 Consequences

Theorem 4.1 (MW Theorem 3). *If ρ_0 extremizes λ_2/λ_1 in $\mathcal{P}_a$, then the cell extension ρ_n of Lemma 2.1 extremizes λ_{2n}/λ_n . Consequently*

$$\mu_{2n,n}(a) = \mu(a), \quad \nu_{2n,n}(a) = \nu(a) \quad (n \geq 1).$$

Proof. If ρ_0 maximizes, then $\nu(a) = \lambda_2(\rho_0)/\lambda_1(\rho_0) = \lambda_{2n}(\rho_n)/\lambda_n(\rho_n) \leq \nu_{2n,n}(a)$ by definition, and Theorem 3.2 gives the reverse inequality; hence ρ_n maximizes λ_{2n}/λ_n . The minimum case is analogous with Theorem 3.5. $\square$

Corollary 4.2 (reduction to the first pair). *For each $0 < a \leq 1$,*

$$\sup_{n \geq 1} \sup_{\rho \in \mathcal{Q}_a} \frac{\lambda_{n+1}(\rho)}{\lambda_n(\rho)} = \nu(a).$$

Here $\nu(a)$ still means the first-pair supremum, not an assumed formula.

Proof. For each fixed n , $\lambda_{n+1} \leq \lambda_{2n}$ and the measurable doubling identity give $\lambda_{n+1}/\lambda_n \leq \nu(a)$. Conversely the outer supremum includes $n = 1$, whose supremum is $\nu(a)$ by definition. No extremizer or explicit first-pair value is needed for this reduction. $\square$

Corollary 4.3 (closed value using the separate first-pair theorem). *Let $0 < \alpha \leq A < \infty$, $R = A/\alpha$, and allow all measurable densities $\alpha \leq \rho \leq A$ a.e. on a finite interval. Using the first-pair variational structure theorem in `SL_ratio_proof.tex`, one obtains*

$$\sup_{n \geq 1} \sup_{\rho} \frac{\lambda_{n+1}}{\lambda_n} = \nu(1/R) = \left(\frac{\arccos(-\sqrt{R}/(\sqrt{R}+1))}{\arccos(\sqrt{R}/(\sqrt{R}+1))} \right)^2.$$

On a unit interval it is attained at $n = 1$ by $[\alpha, A, \alpha]$ with relative widths st, t, st , $s = \sqrt{R}$, $t = 1/(2s+1)$.

Proof. Divide the density by A to obtain the normalization $[1/R, 1]$, and normalize the interval affinely. Both operations preserve ratios. Corollary 4.2 gives the supremum as $\nu(1/R)$. The companion’s theorem entitled “first-pair global extremal structure” proves global optimality by compactness, first variation, a zero-energy identity, and interface matching; its balanced-phase calculation then evaluates that optimizer. Neither of those two first-pair steps uses a doubling identity or this corollary. Thus this reference has one direction of dependence and supplies the missing upper bound, not merely a candidate value. The case $R = 1$ is the constant-density spectrum and has value 4. $\square$

5 Source locations and proof dependencies

The local primary sources are `papers/mw1976.pdf` and `papers/keller1976.pdf`, relative to the project root. Their text extractions serve only as search indexes; formulas and the cited statements must be checked on the PDF pages. In [1], printed pp. 517–518 (PDF pp. 1–2) define the original class as piecewise continuous densities with a bounded number of jumps and $0 < a \leq \rho \leq 1$. Theorems 1–2 on p. 518 state the two-jump and symmetry properties, and Theorem 3 on pp. 519, 529 states the extremizer transfer. The cell construction is in Lemma 1, pp. 521–523, with the distinct factors $(-s)^j$ and t^j in (5.3)–(5.4); Lemma 2, pp. 523–524, gives the reverse inequalities by truncation. Our signed b_1, b_2 convention makes these two signs explicit. The original only sketches the minimum case; the three-case proof above includes its common-zero branch.

Keller’s first-variation and switching formulas are on printed pp. 486–487, (3.1)–(3.9); his p. 488 explicitly attributes the two-jump symmetry result to Mahar–Willner. These locators distinguish the historical results from the proofs supplied here. The present doubling proof needs the stated classical spectral and Sturm facts, not the companion’s first-pair extremal theorem. The latter enters only Corollary 4.3. The extension to $\mathcal{Q}_a$ is justified by Proposition 1.1, not by silently enlarging the original papers’ density class.

No existence theorem for higher-index extremizers is used in the truncation inequalities: each is proved for an arbitrary density before taking its extremum. No assertion about fixed $n \geq 2$ adjacent-ratio optimizers or any spectral-gap extremum follows from these lemmas.

6 Historical numerical records and current checking limits

The preceding version reported transfer-matrix checks of cell extension, truncation and zero motion, under the script names `op05_mw_lemma12_audit.py`, `op05_mw_lemma2_audit.py`, `op05_mw_mincase_audit2.py`, and `op05_mw_extra_checks.py`. Those historical runs have not been re-executed as part of this author revision and are not evidence for the new proofs. In particular, random strict-inequality cases cannot check the omitted equality case. Constant densities give that case for every $m \geq 2$, as shown explicitly above. Current author checks and their precise limits are recorded in the external `ratio-author.md`; independent acceptance and PDF builds are left to the coordinator.

References

- [1] T. J. Mahar, B. E. Willner, *An extremal eigenvalue problem*, Comm. Pure Appl. Math. 29 (1976), no. 5, 517–529. <https://doi.org/10.1002/cpa.3160290505>
- [2] J. B. Keller, *The minimum ratio of two eigenvalues*, SIAM J. Appl. Math. 31 (1976), no. 3, 485–491. <https://doi.org/10.1137/0131042>